FNB HACKATHON · 2026

TrustKhaya



Your informal financial life, made visible.

One explainable Community Trust Score. One capped community guarantor.
A zero-fee FNB account, decided by a human.

Live demo: **graffgrid.co.za/work/trustkhaya** Video: **86-second system demo** Demo PIN **1234**

Built by **Mpho Dlamini**

THE PROBLEM

The first bank account is gated by paper that millions of young South Africans do not have.

No payslip

Work is real, but informal. No employer letter to staple to a form.

No utility bill

They live at home. Nothing arrives in their own name.

No credit history

No bureau file, so the risk models have nothing to read.

But the behaviour is there, and it is disciplined: twenty-four months of stokvel contributions, a wallet that receives money every week, tuck-shop credit repaid on time, and an elder who will stand behind them. None of it reaches a credit bureau.

What we measure instead

Observed consistency with informal financial commitments — not character, not income.

THE SOLUTION

A score the bank can underwrite, a guarantee with a limit, and an explanation a person can read.

1 · Score

One hundred bricks

Stokvel 35, wallet 25, spaza 20, guarantor 20. Each part is a wall of the same house; each brick is one point, computed live from raw rows.

2 · Guarantee

Capped, not a referral

A guarantor picks R500, R1 000 or R2 000 — the most they could ever be asked to cover. Six rules cap the total at R5 000 across three people.

3 · Decision

Explained, then approved

Every point is attributed in plain language. The underwriter sees the delta, the guarantee and the audit trail, and a human makes the call.

Outcome: a zero-fee first account for someone with no payslip, no bill in their name and no bureau file — with the risk shared, bounded, and visible.

75 → 91 with one R1 000 vouch

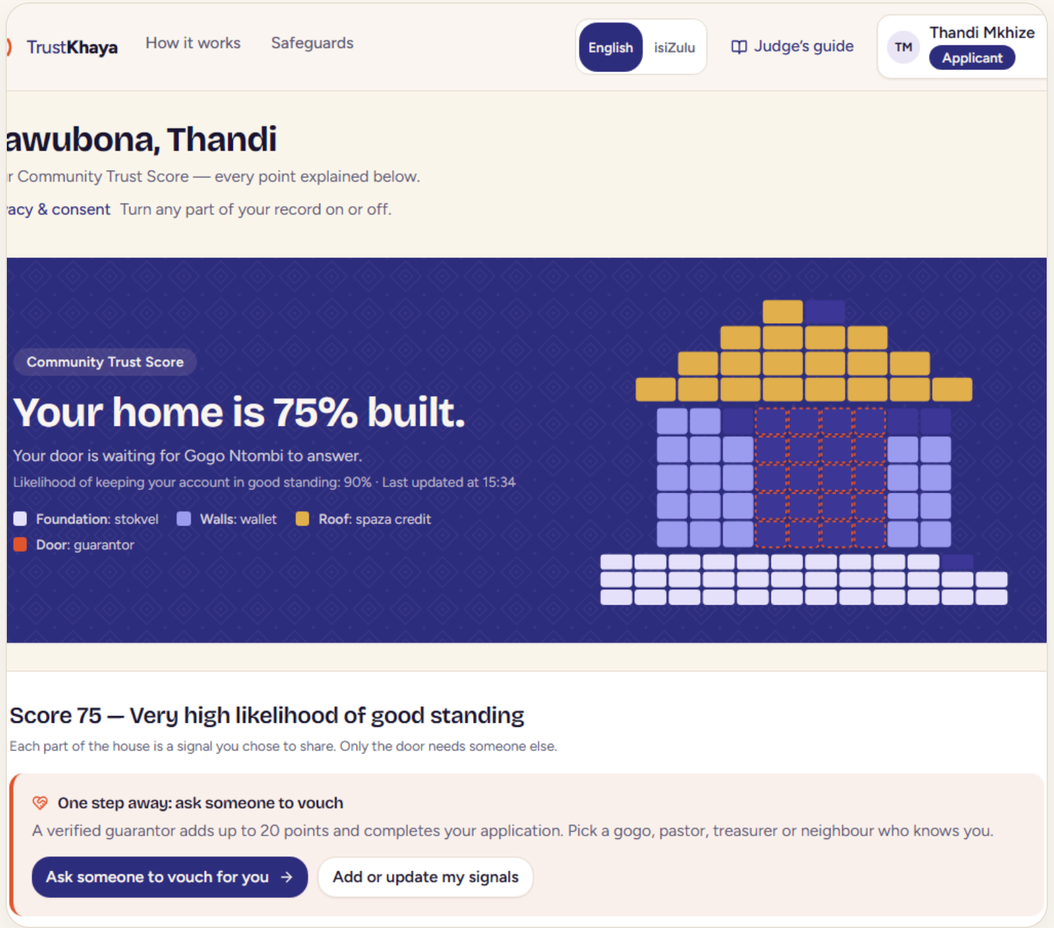
HOW THE SCORE WORKS

Four signals, one house, no black box.

PART	SIGNAL	MAX	WHAT FILLS IT
Foundation	Stokvel	35	tenure and consistency
Walls	Wallet	25	stability, regularity, tenure
Roof	Spaza credit	20	on-time rate, 3-day grace
Door	Guarantor	20	verified only; 0.5 / 0.8 / 1.0

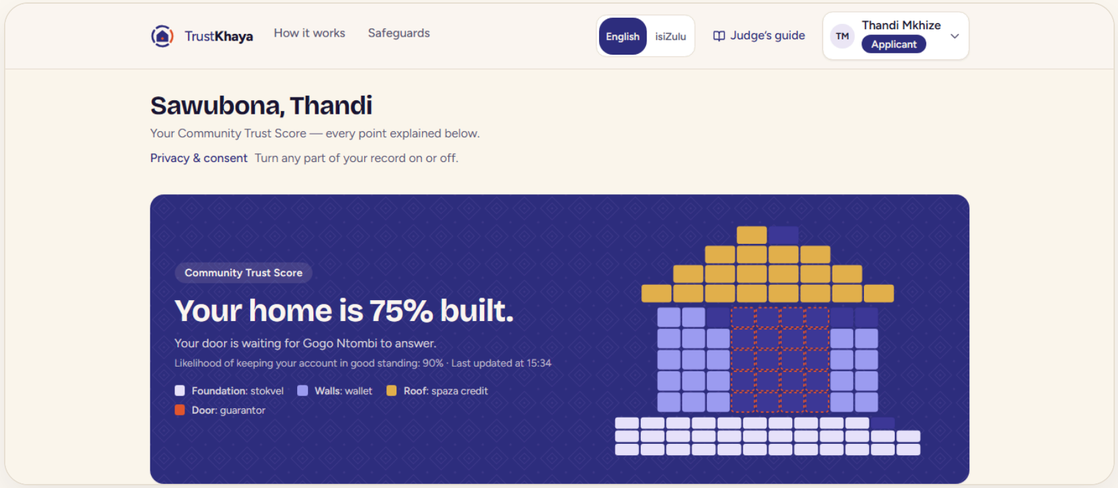
```
guarantor = { R500: 0.5, R1000: 0.8, R2000: 1.0 }
z = −3.0 + 4.2·consistency − 2.0·walletCv
    + 1.5·spazaOnTime + 1.0·guarantorFactor
probability = 1 / (1 + e^−z), clamped 5–97%
```

Bands: Strong ≥ 70, Building 40–69, Early days < 40. Approve needs 70+, a verified guarantor and 75%+; the score without the guarantee is shown rebased out of 80.

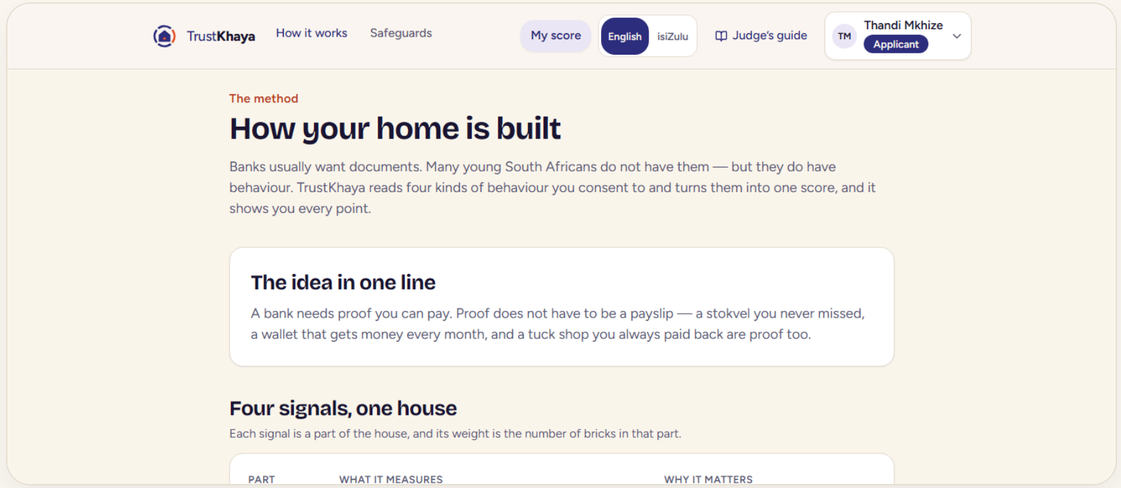


THE PRODUCT

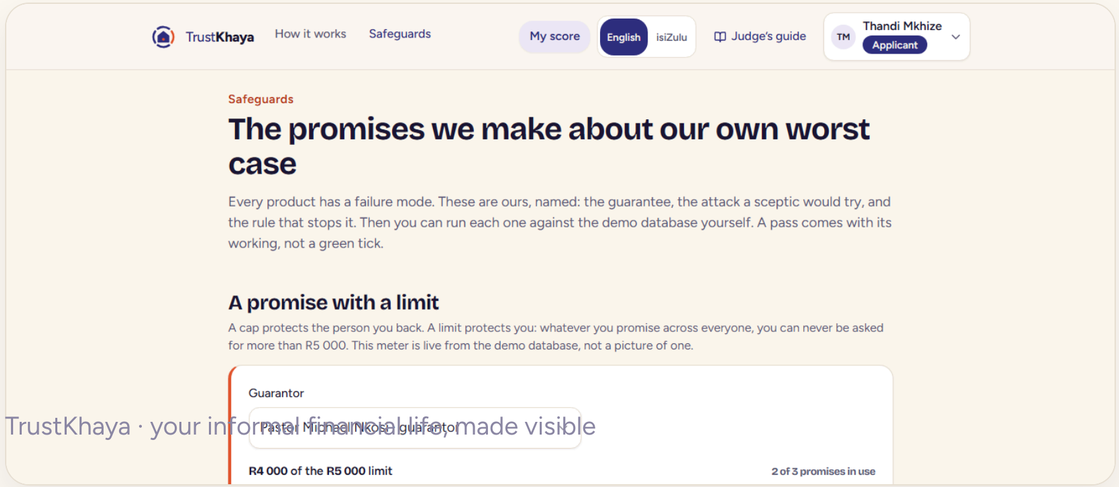
Built, not mocked: four working surfaces.

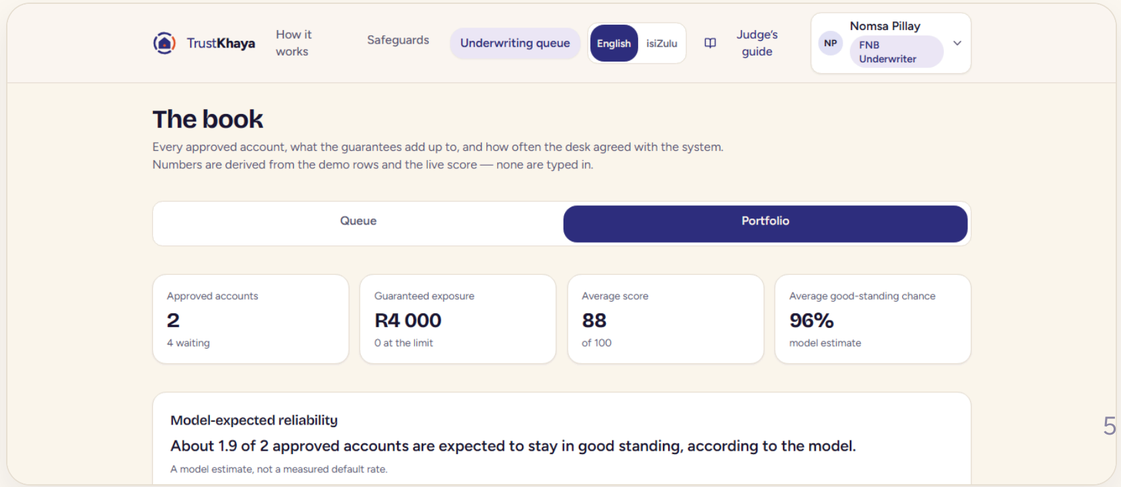


Applicant dashboard — the score, explained



The method — weights, formulas, worked example





THE USER JOURNEY

Fifty seconds from an invisible file to an approved account.

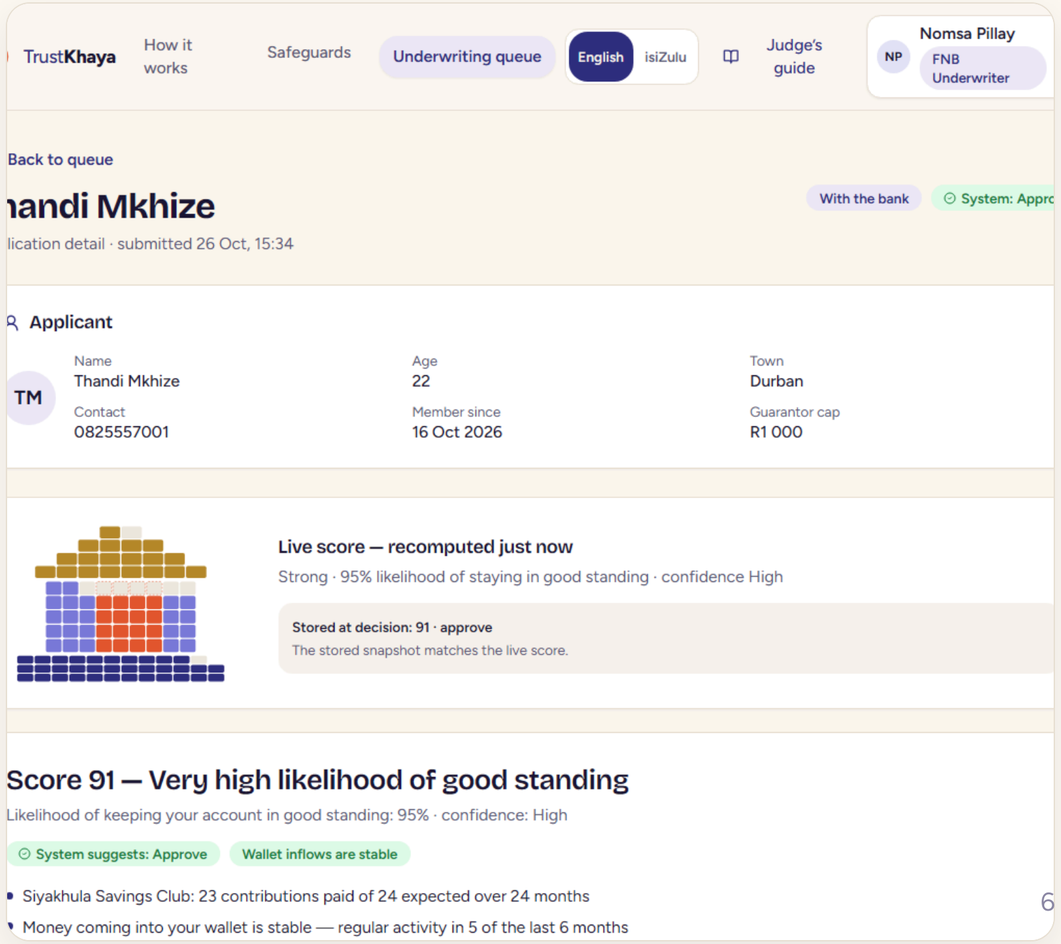
1. **Thandi signs in**
   22, Durban. No payslip, no bill, no bureau file. Her score is 75 — Strong, but the door is unlit.

2. **She asks Gogo Ntombi**
   A vouch link. Gogo sees first name, town, age band, score band and qualitative chips — never an amount.

3. **Gogo signs R1 000**
   Six rules run server-side before anything is written. Only the door lights: +16 points, 91/100, Approve.

4. **FNB decides**
   The underwriter sees the delta (94 without the guarantee), the guarantor's exposure, the audit trail — and approves with a written reason.

5. **Zero monthly fee**
   Thandi sees the decision, the three next steps, and a stokvel debit order to keep the habit going.



TECHNICAL ARCHITECTURE

One pure engine, one seam to the database, one honest UI.

Browser · PWA
Next.js 14 App Router, React 19, TanStack Query, Tailwind, installable service worker

→

API routes
/api/* route handlers, zod on every input, signed httpOnly session cookie, persona-based authorisation

→

lib/repo.ts — the single seam
Every database call lives here. Swapping Supabase for another store touches one file.

→

Supabase · Postgres
Ten tables, RLS deny-all for the anon key, service-role access server-side only

lib/scoring.ts — pure, deterministic, tested

No clock, no randomness, no I/O. The same rows always produce the same score, which is why the house, the explanation, the what-if preview and the underwriter's snapshot can never disagree.

Verification

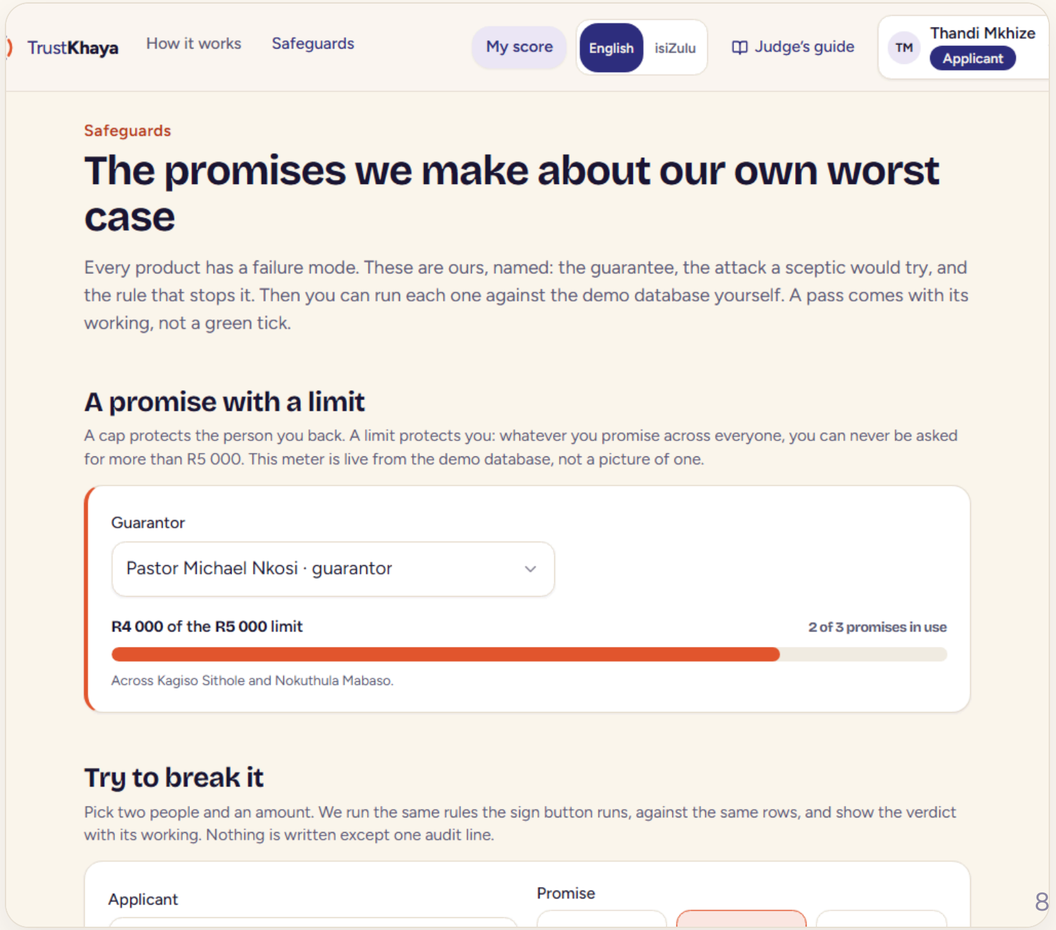
164 tests across the engine, the house geometry, the safeguards, the vouch rules, consent, portfolio maths, history and i18n. **tsc** clean, **lint** clean, production build green.

TRUST, BY CONSTRUCTION

A guarantee is only credible if it is limited, visible and enforced.

RULE	VERDICT
Nobody vouches for themselves (id, phone or email)	Blocked
One verified vouch per applicant	Blocked
At most three people backed at once	Blocked
Total promise never exceeds R5 000	Blocked
Three promises is flagged to the bank	Warn
Two people naming each other is flagged	Warn

Consent is granular and withdrawable: stokvel, wallet, spaza, share with guarantor, share with the bank. Withdrawing a signal removes it from the score at read time — and the page shows the exact drop before you do it.



SEE IT WORKING

Sixty seconds, no setup.

LIVE DEMO

graffgrid.co.za/work/trustkhaya

The interactive slice runs the real engine in your browser. The full Next.js product is the system in the video.

DEMO ACCOUNT

+27 82 555 7001 · PIN 1234

Thandi Mkhize (applicant). Every seeded persona uses PIN 1234 — Gogo Ntombi, Pastor Michael, Nomsa Pillay (FNB).

DEMO VIDEO · 86 SECONDS

graffgrid.co.za/work/trustkhaya/demo/TrustKhaya_Functional_Demo.mp4

1080p, narration, web + mobile side by side.

TRY THIS

1 · Open Thandi → see 75 and the unlit door
2 · Guarantor tab → sign R1 000 → the door lights
3 · FNB desk → approve → back to Thandi, approved

WHAT IS HONEST ABOUT THIS

Simulated data, real mechanics, stated limits.

What is real

- The scoring engine, the house, the explanation and the recommendation rules
- The vouch safeguards, enforced server-side before any write
- Consent filtering, the audit trail, the underwriter decision and its snapshot
- 164 tests, a clean typecheck, a clean lint, a production build

What is not

- All signals are simulated — no bureau, wallet provider or bank is contacted
- The guarantor agreement is a demo, not a legally binding contract
- Auth is demo-grade: a signed cookie and a shared PIN, no rate limiting
- Real deployment needs POPIA/DPIA review, NCA compliance and fairness testing

The pitch in one line: social collateral, priced and bounded, turns an invisible financial life into a first bank account.

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